ChemCritic 文献证据卡

PMC12933482 | 不对称氢化 | 采集状态 imported_unchecked

标题	Iridium-Catalyzed Regio- and Enantioselective Hydrogenation of 5 - Alkylidene Cyclopentenones
作者	Wei H., Ren J., Yuan Q., Deng W., Zhang W.
出处	Precision Chemistry 2025
DOI	10.1021/prechem.5c00087
原文链接	https://pmc.ncbi.nlm.nih.gov/articles/PMC12933482/
许可	https://creativecommons.org/licenses/by-nc-nd/4.0/This article is licensed under CC-BY-NC-ND 4.0

摘要（原文节选）

The regio- and enantioselective hydrogenation of the C C bond in molecules with two or more C C bonds of the same type is very challenging but highly desirable. Herein, we report a regio- and enantioselective hydrogenation of 5-alkylidene cyclopentenones catalyzed by an Ir/BiphPHOX complex, delivering the corresponding mono- and double-hydrogenated products in excellent results, respectively (5-substituted cyclopentenones: up to 99% yield and 98% ee; 2,4-disubstituted cyclopentanones: up to 99% yield, 16:1 dr, and ⋯

数据表 · 1（7 行，含数值行 7 行）

Reaction Condition Screening

entry	solvent	L	conv. (%)	2a:3a	ee (%)
1	DCM	L1	99	95:5	98
2	DCM	L2	93	95:5	96
3	DCM	L3	99	95:5	97
4	DCM	L1	96	95:5	98
5	DCM	L1	81	>99:1	98
6	DCM	L1	99	>99:1	98
7	o-xylene	L1	41	>99:1	97

脚注（逐字）：aReaction conditions: 1a (0.2 mmol) and [Ir(L)COD]BArF (1 mol %) in DCM (2 mL) under H2 (20 bar) at room temperature for 24 h.bDetermined by 1H NMR analysis of the crude reaction mixture.cDetermined by chiral HPLC.d12 h.e10 ° C.fH2 (50 bar), 10 ° C. BArF = tetrakis[3,5-bis(trifluoromethyl)phenyl

正文对应句（逐字）：Reaction conditions: 1 (0.2 mmol) and [Ir(L1)COD]BArF (1 mol %) in DCM (2 mL) under H2 (50 bar) at 10 ° C for 24 h. Isolated yield.

数据表 · 3（5 行，含数值行 5 行）

Optimization of the Double Hydrogenation Reaction Conditions

entry	substrate	conv. (%)	yield (%)	dr	ee (%)
1	1a, R2 = Me	>99	3a, 8%	–	–
2	1a, R2 = Me	>99	3a, 99%	1:1	95/98
3	4a, R2 = MeO2C	>99	5a, 99%	12:1	99
4	4b, R2 = PhC(O)	>99	5b, 99%	4:1	95
5	4a, R2 = MeO2C	>99	5a, 99%	12:1	99

脚注（逐字）：aReaction conditions: 1 or 4 (0.2 mmol), [Ir(L1)COD]BArF (5 mol %) in DCM (2 mL) under H2 (50 bar) at 60 ° C for 24 h.bIsolated yield.cDetermined by 1H NMR analysis of the crude reaction mixtures.dDetermined by chiral HPLC.e[Ir(L1)COD]BArF (1 mol %) was used.fIt is difficult to determine which

正文对应句（逐字）：Reaction conditions: 1 (0.2 mmol) and [Ir(L1)COD]BArF (1 mol %) in DCM (2 mL) under H2 (50 bar) at 10 ° C for 24 h. Isolated yield.

本卡为 NCBI PMC 开放获取 XML
的逐字摘录（标题、作者、表格行列、脚注、正文句均未改写），用于参照集候选与覆盖面登记；未逐条人工核验，标记 imported_unchecked，不写入判定记录、不参与 ChemCritic 引擎的可比性判定。对外引用一律回到 DOI 原文核对页码与表号。